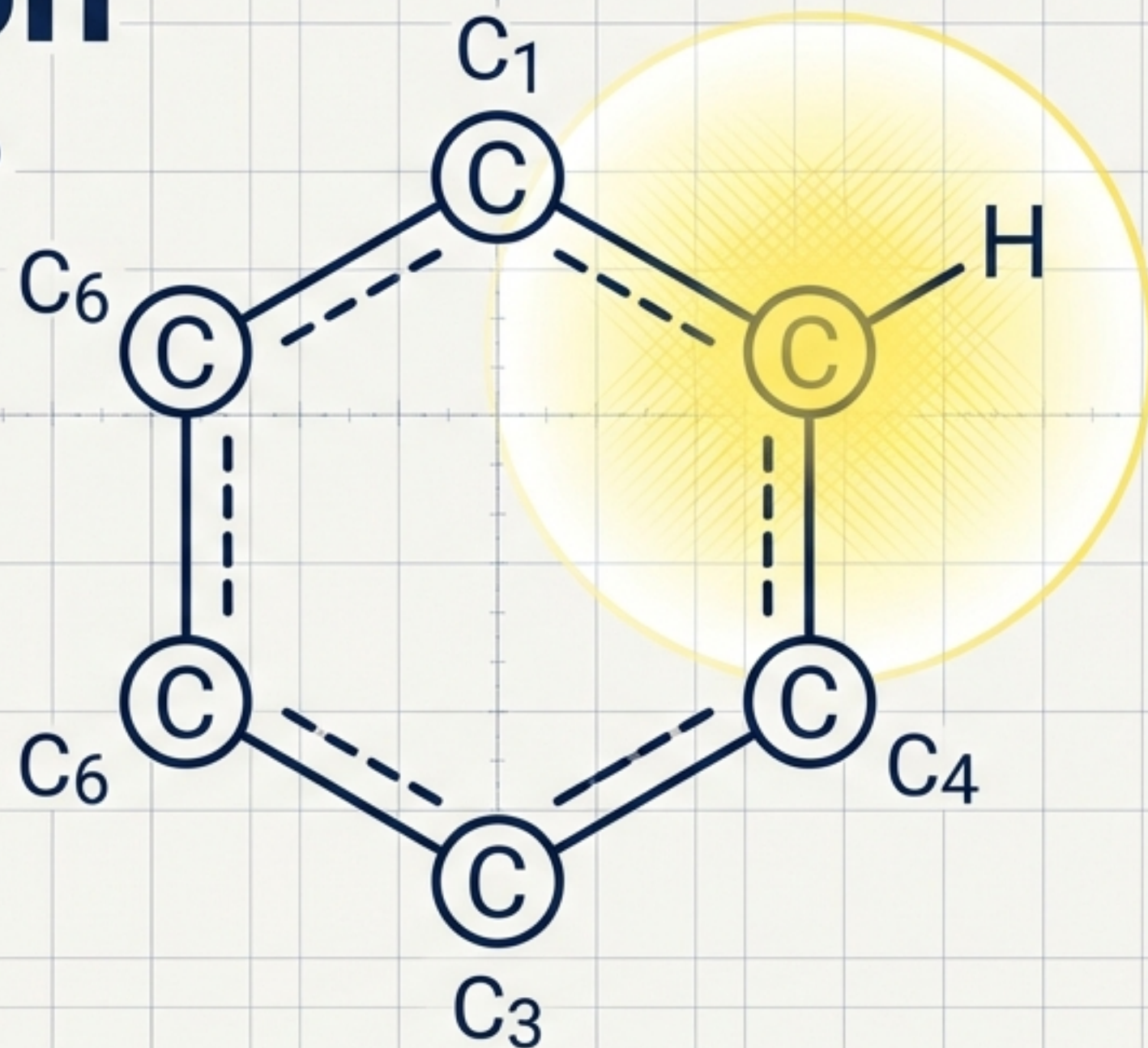


Decoding Aromatic Substitution

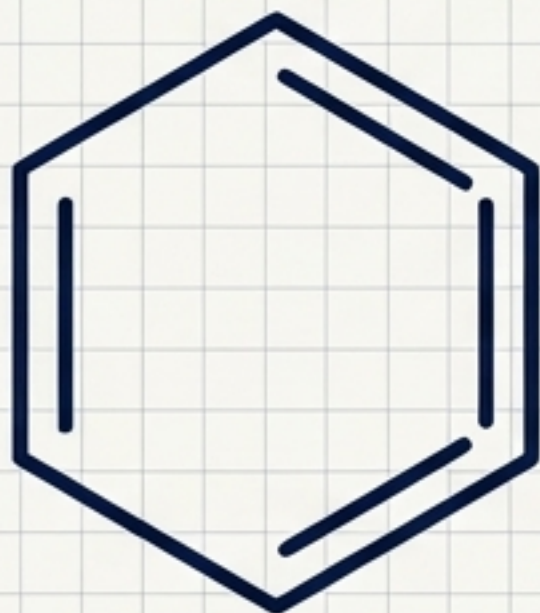
From Empirical Mystery to
First-Principles Mastery



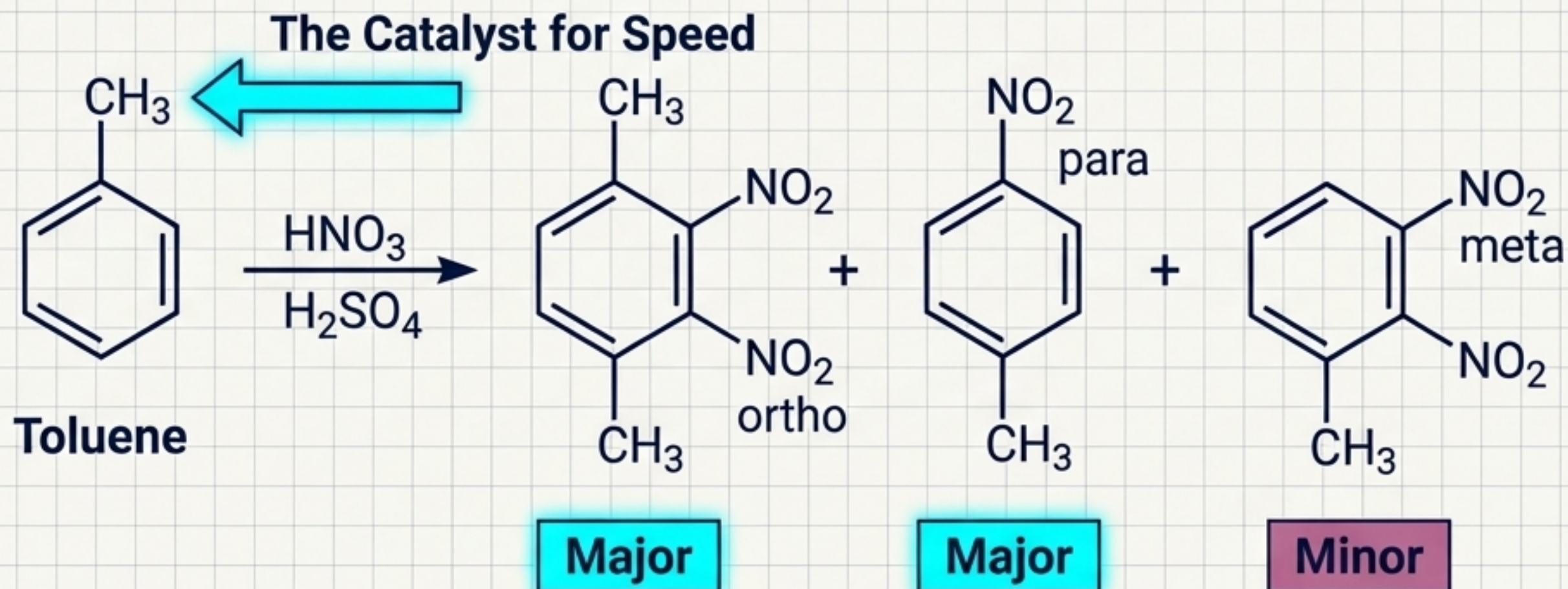
The 25x Speed Mystery

When toluene and benzene are exposed to nitric and sulfuric acids under identical conditions, toluene reacts **25 times faster**. Furthermore, the incoming nitro group doesn't land randomly; it **targets** specific physical locations (**ortho** and **para**) on the ring.

BENZENE REFERENCE



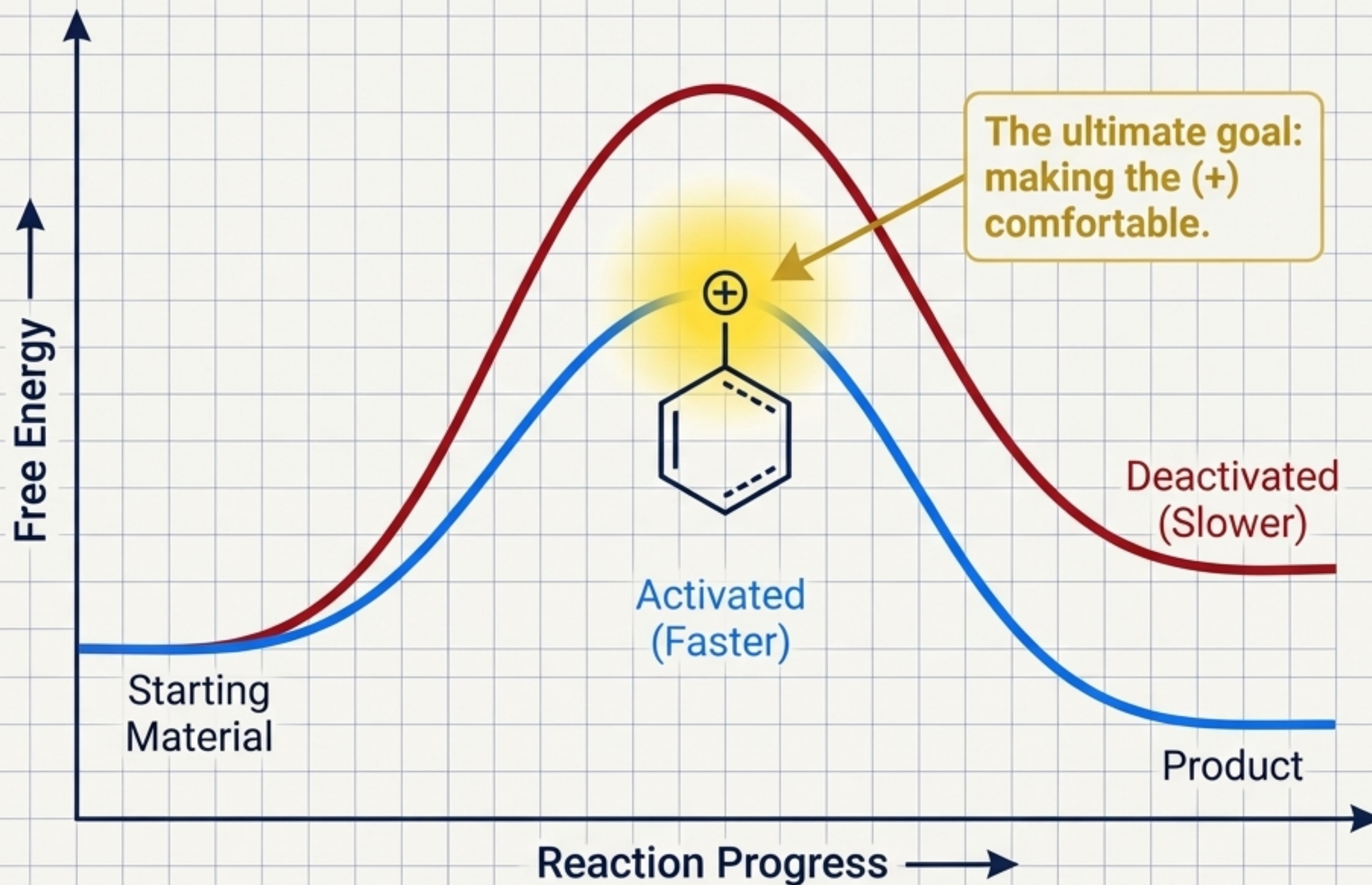
TOLUENE REACTION



The Universal Anchor: Carbocation Stability

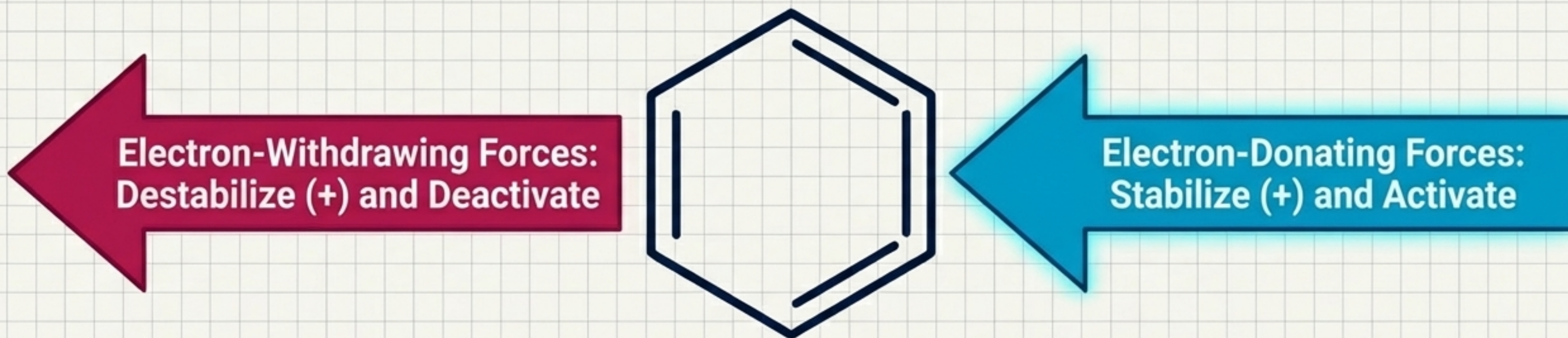
The addition step—generating the carbocation—is the rate-determining step in electrophilic aromatic substitution.

According to the Hammond postulate, any substituent that stabilizes this positive (+) intermediate lowers the activation energy, accelerating the reaction.



The Invisible Tug-of-War

Substituents manipulate the roaming (+) charge through two distinct invisible forces. The ultimate reactivity and orientation of the ring depend entirely on which of these forces dominates.



Inductive Effect:

Pulling or pushing through the framework of σ -bonds.

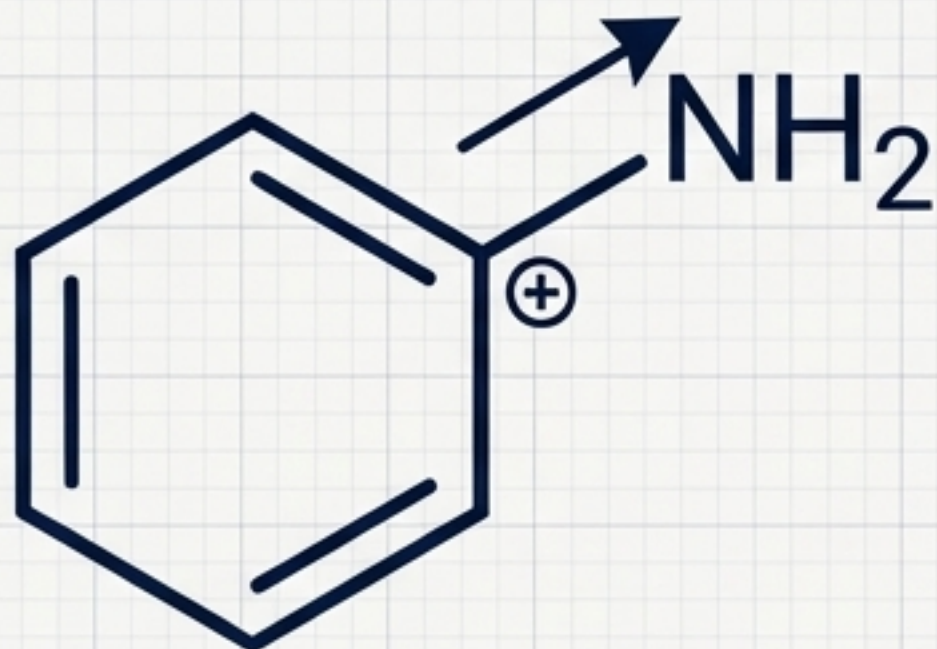
Mesomeric (Resonance) Effect:

Pulling or pushing through the π -system.

Force 1: The Inductive Effect

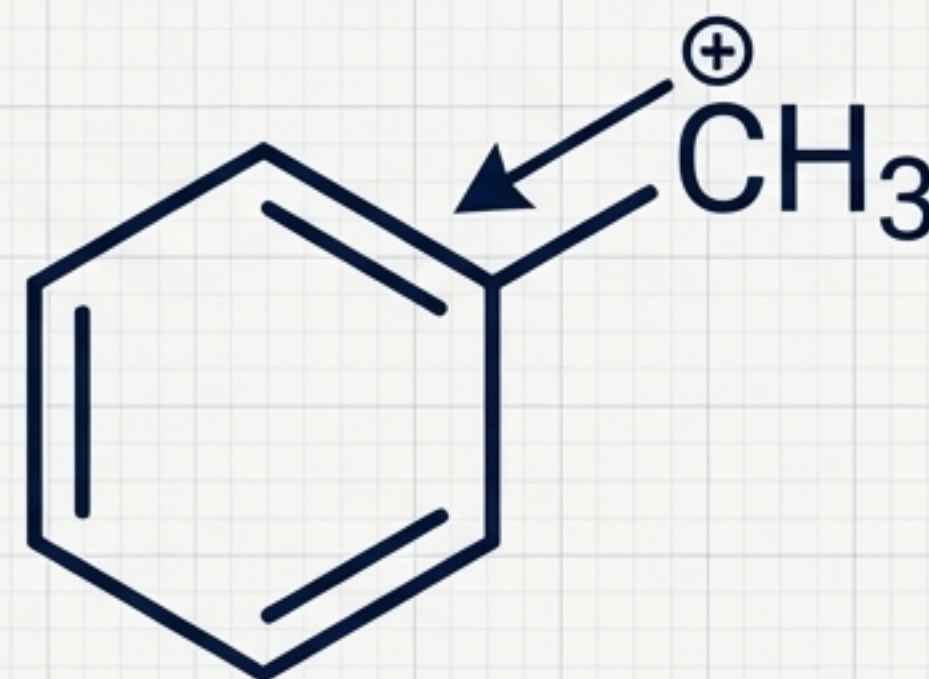
The Inductive Effect operates through single bonds, driven by the physical properties of the attached atoms. Electronegative atoms pull electron density away (-I), while polarizable alkyl groups push electron density toward the ring (+I).

Electron-Withdrawing (-I)



-COOH, -CHO, -NO₂, -CN, -OH, -NH₂, -X

Electron-Donating (+I)



-R (alkyl), -Mg

Force 2: The Mesomeric Resonance Engine

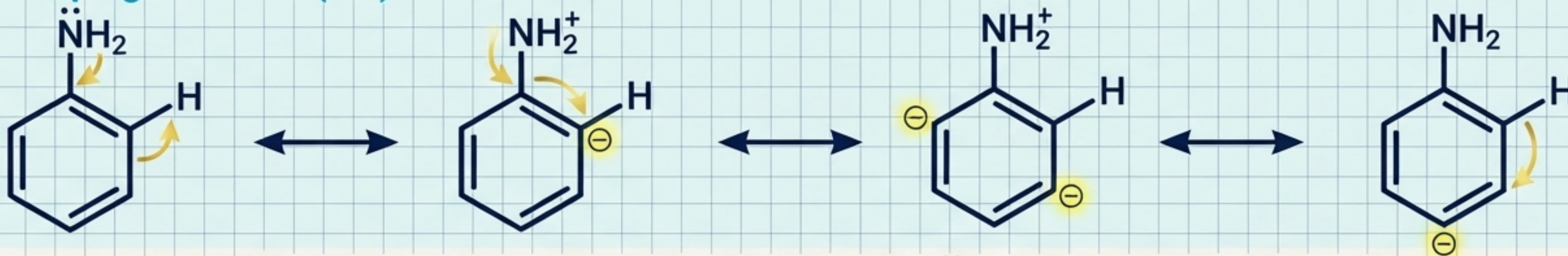
The Mesomeric Effect operates by delocalizing π -electrons and lone pairs through conjugated double bonds. It can either pump electrons into the ring (+M) or siphon them out (-M).

Siphoning Electrons (-M)



Three resonance structures place a destabilizing (+) charge on atoms in the ring.

Pumping Electrons (+M)



Three resonance structures place a protective (-) charge on atoms in the ring.

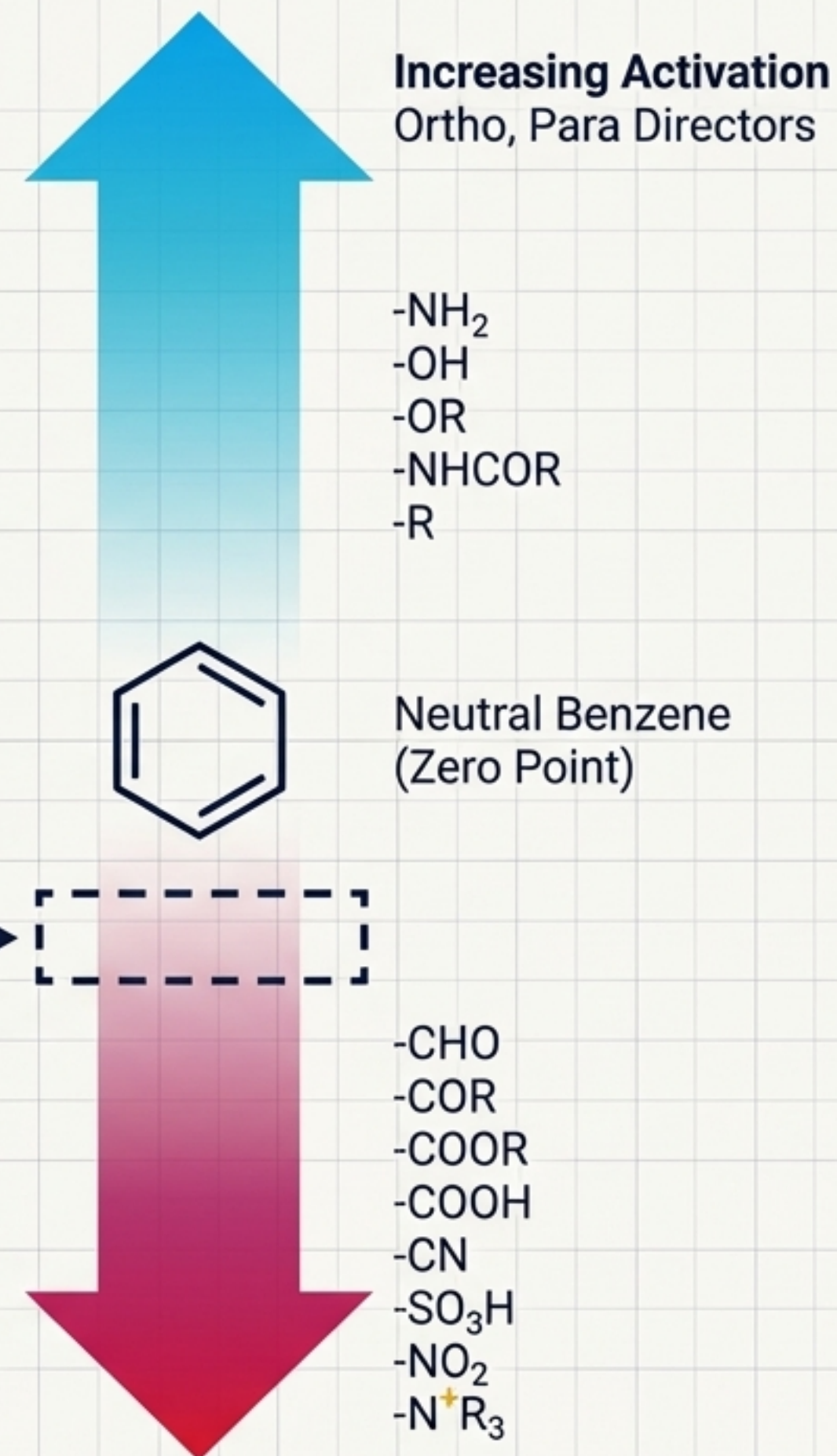
The Reactivity Thermometer

The competing Inductive and Mesomeric forces generate a continuous spectrum of reactivity relative to neutral benzene.

Activators push electron density in;
Deactivators pull it out.

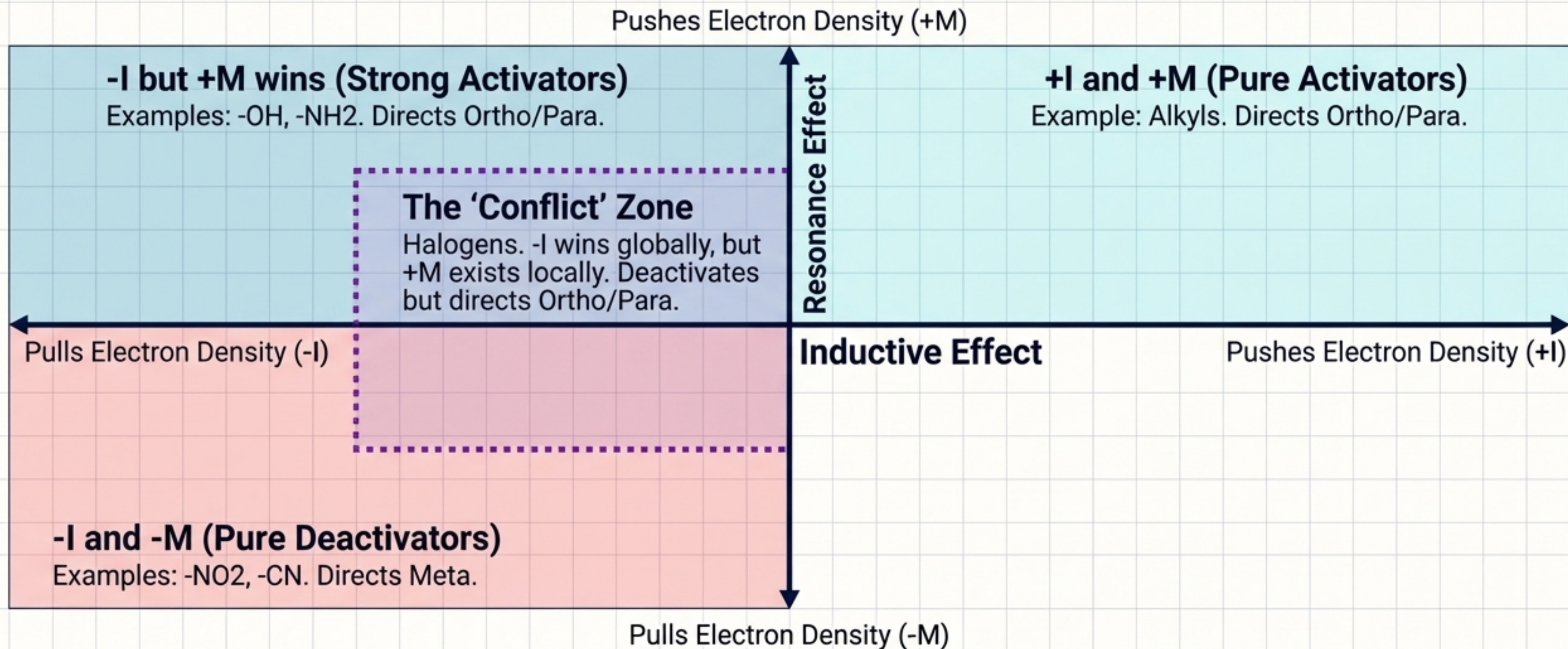
The Exception:
Deactivating but
Ortho/Para Directing.
-X (F, Cl, Br, I)

**Increasing
Deactivation**
Meta Directors



The Directing Group Diagnostic Matrix

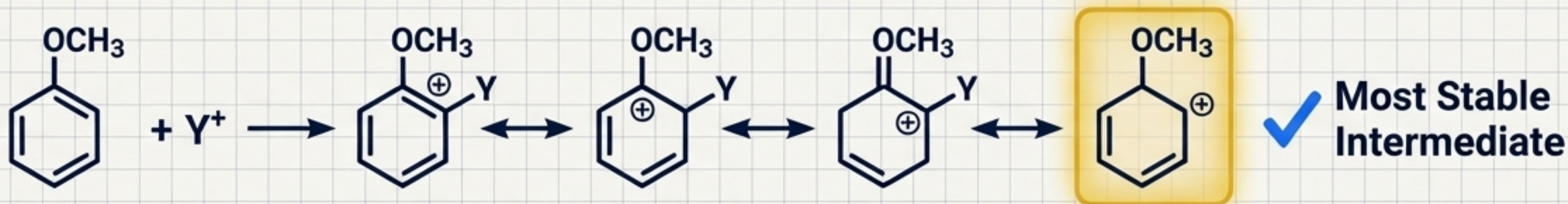
Why do groups behave the way they do? It comes down to which force wins the tug-of-war.



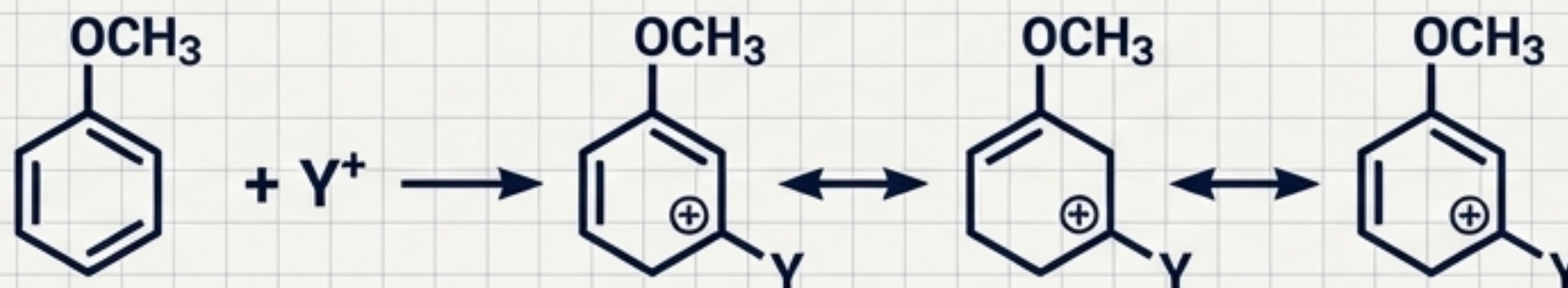
The Ortho/Para Advantage

When an electrophile attacks the ortho or para positions of an activated ring, the resonance engine places the roaming (+) charge directly adjacent to the electron-donating group, achieving maximum stability.

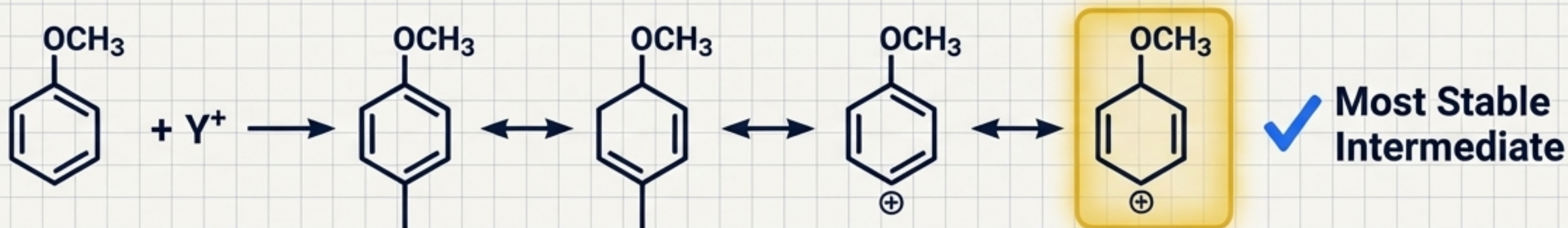
Ortho Attack



Meta Attack

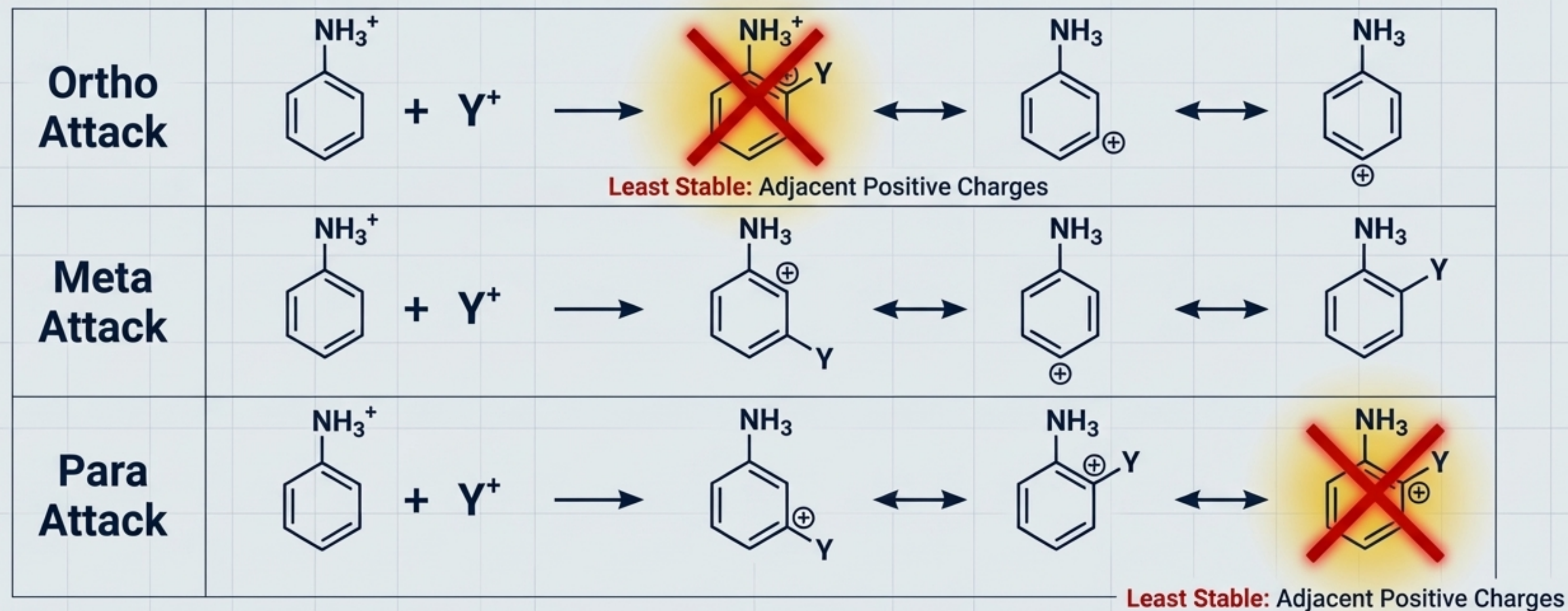


Para Attack



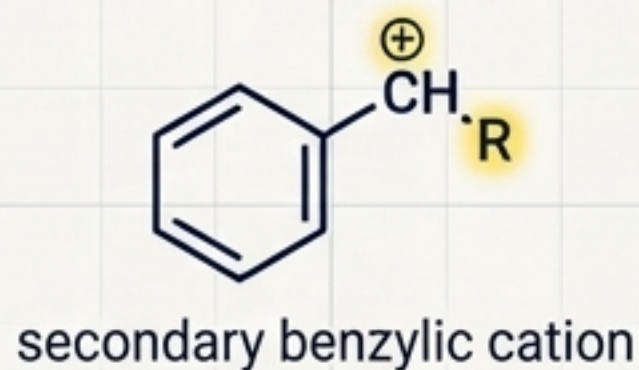
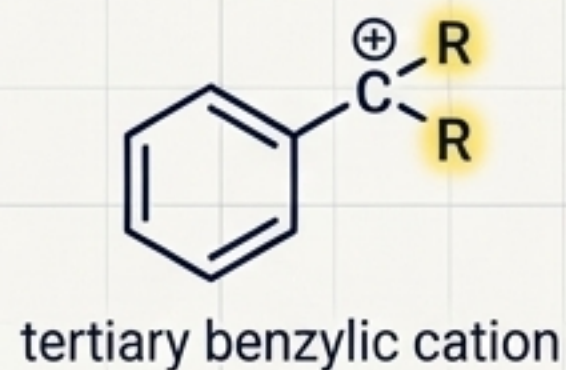
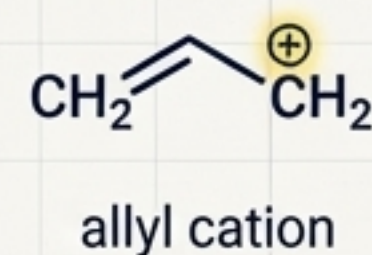
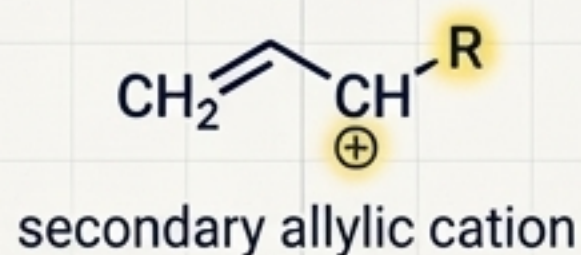
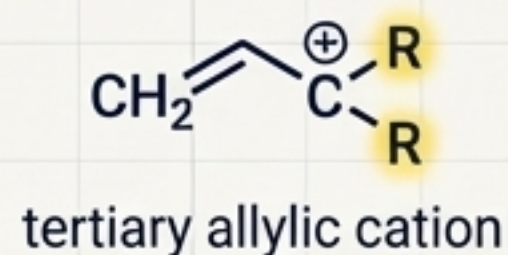
The Meta Compromise

Deactivating groups don't stabilize the meta position; they actively destroy the ortho and para positions. Ortho/para attacks force the (+) charge directly next to the electron-withdrawing group, creating catastrophic instability.



Expanding the Resonance Engine: Benzylic Cations

The stability-reactivity principle extends to side chains. A **benzylic cation** shares its positive charge across four different carbons, dramatically increasing stability through extreme delocalization.



Decreasing Stability

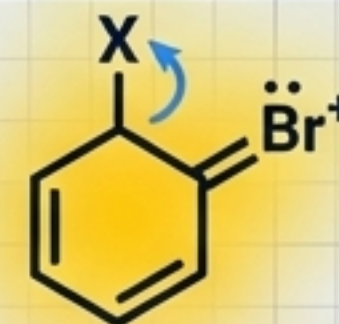
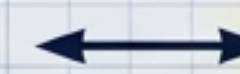
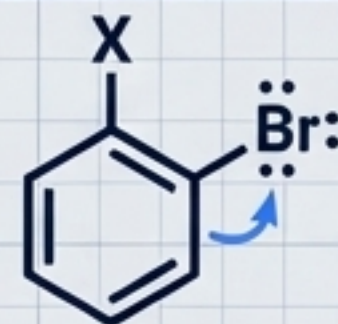
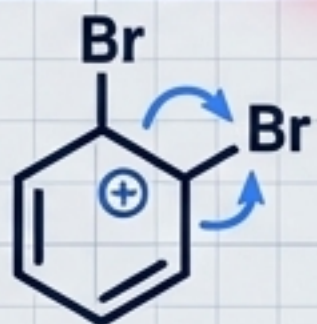
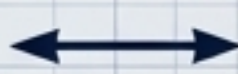
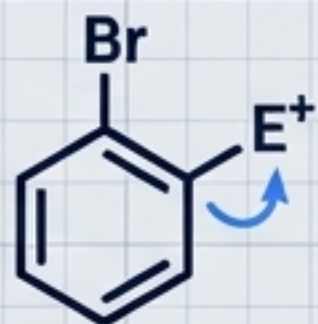
The Halogen Paradox

Halogens break the binary rules. Because their **electronegativity (-I)** outweighs their **resonance (+M)**, they **globally drain electron density**, deactivating the ring. However, their lone pairs still provide precise +M stabilization **only** at the ortho and para positions.



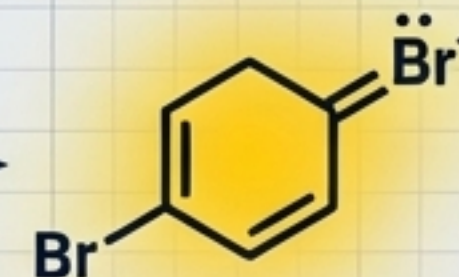
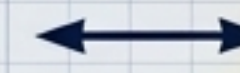
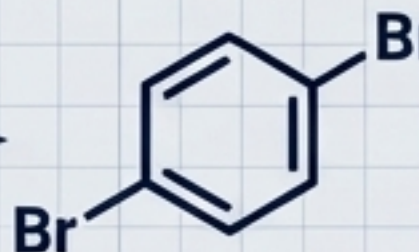
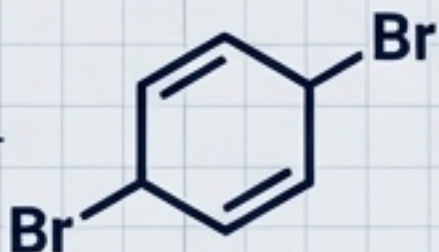
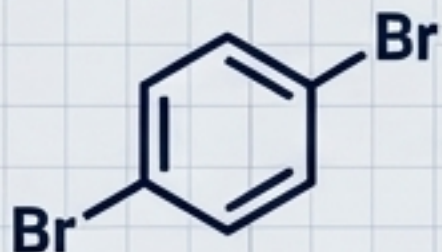
Global Deactivation ($-I > +M$)

Ortho Attack



Local Ortho/Para Rescue (+M)

Para Attack



Local Ortho/Para Rescue (+M)

The Aryl Halide Fortress

Unlike alkyl halides, typical aryl halides are highly unreactive toward S_N1 and S_N2 nucleophilic substitution. Their physical architecture creates an impenetrable barrier.

1. The Steric Shield:

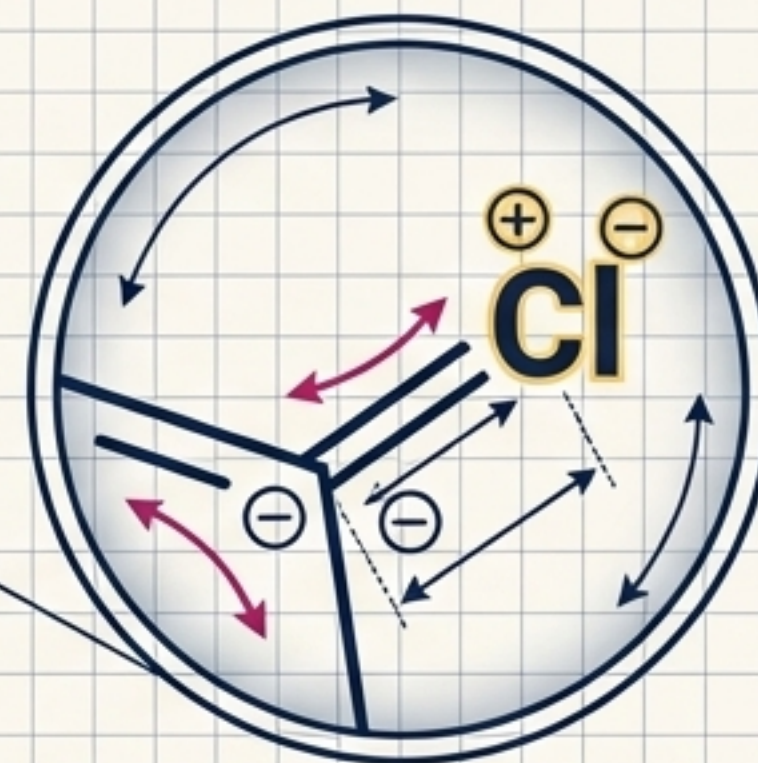
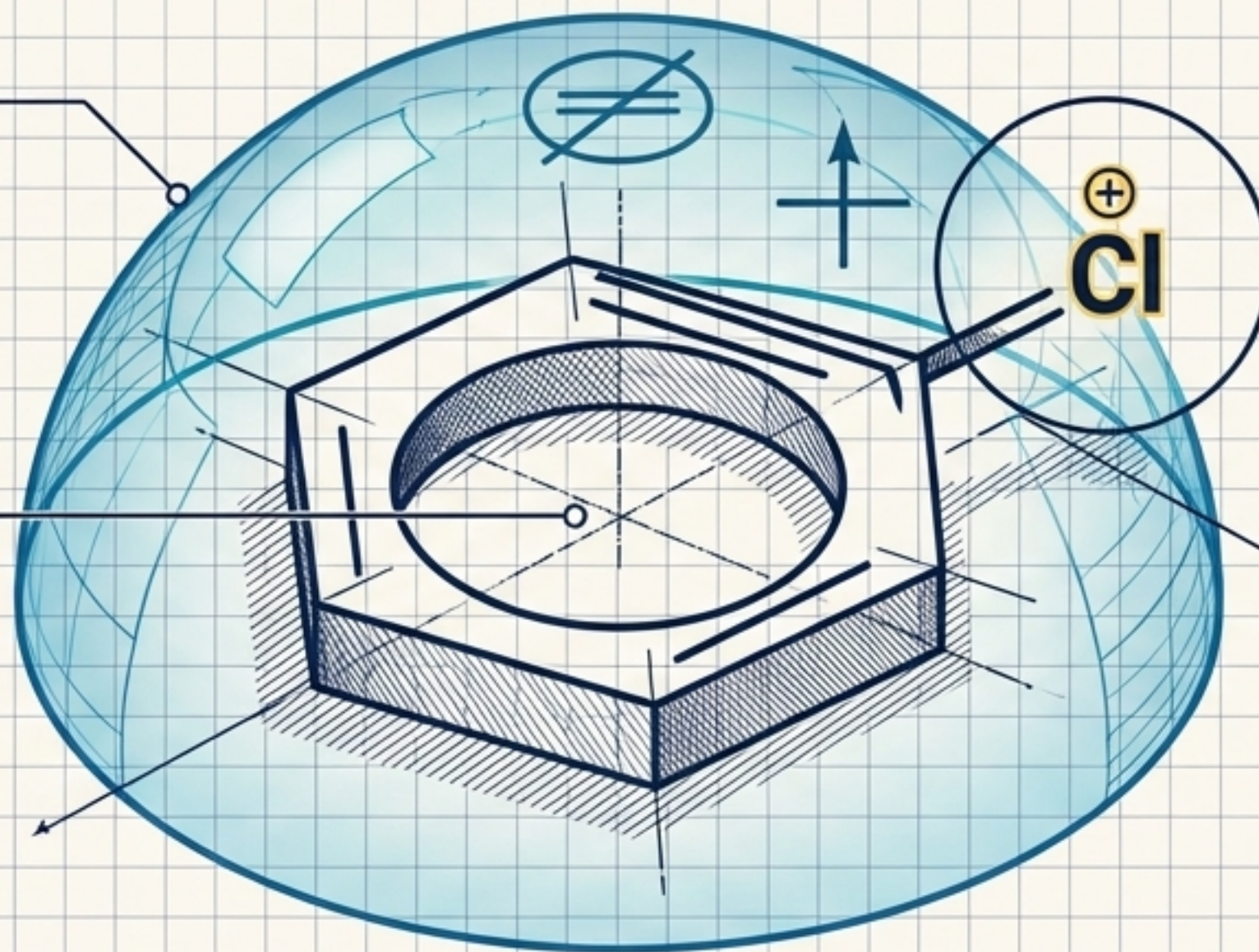
Bulky benzene ring physically blocks backside S_N2 attack.

2. The Unstable Core:

Unstable phenyl cation intermediate makes S_N1 substitution impossible.



Unstable Intermediate



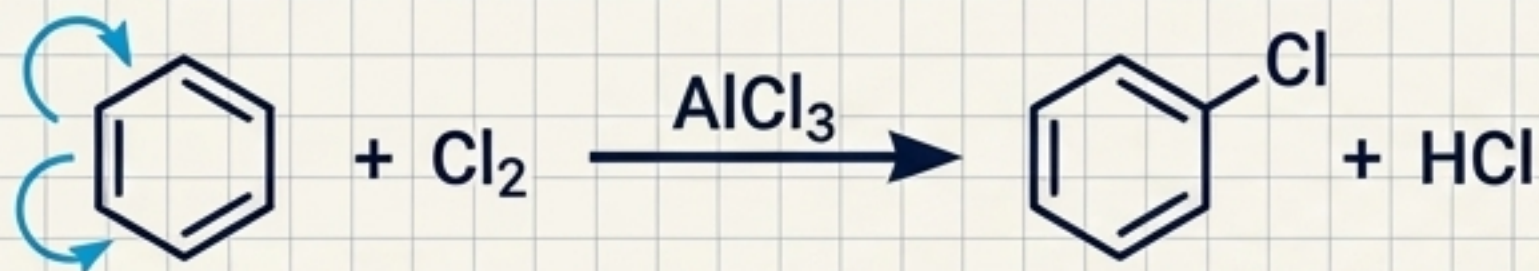
3. The Iron Grip:

sp^2 hybridization and partial double-bond character make the bond shorter and stronger than alkyl bonds.

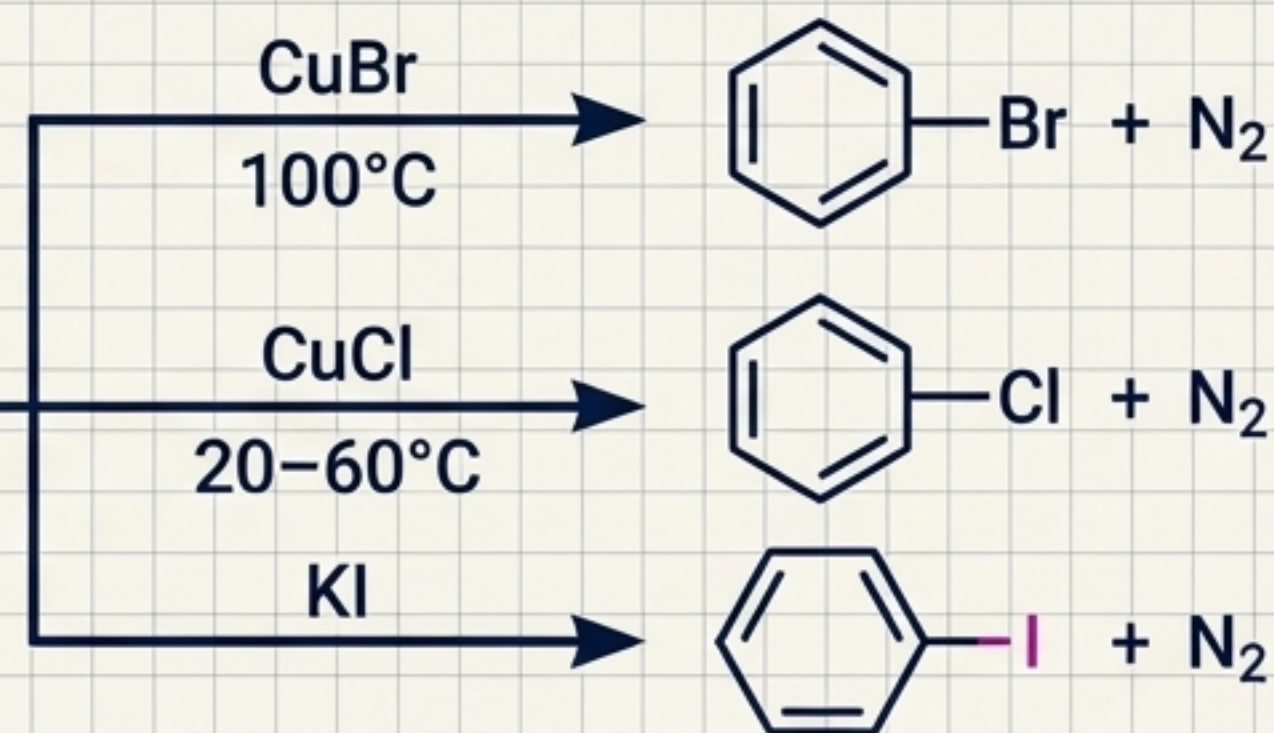
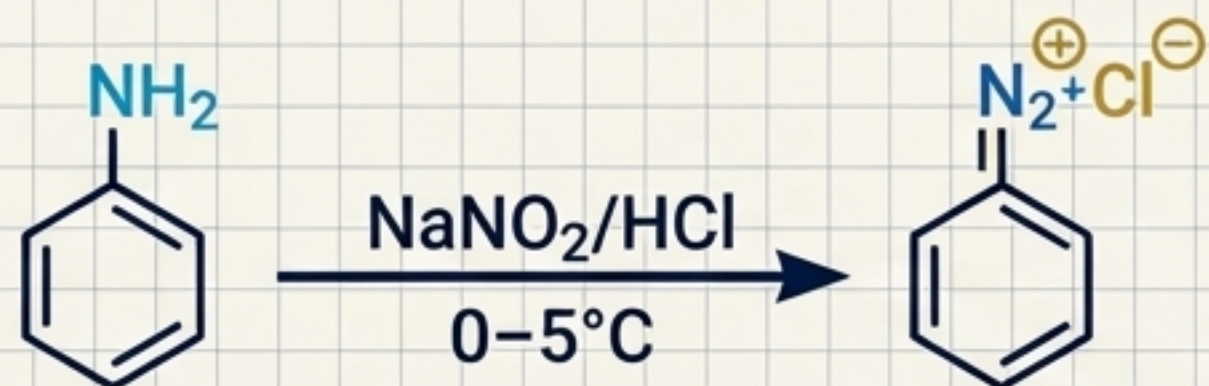
Constructing the Fortress: Synthesis

Because aryl halides resist substitution once formed, creating them requires specialized pathways: direct halogenation or the intermediate use of diazonium salts.

Path 1: Halogenation (Friedel-Crafts)

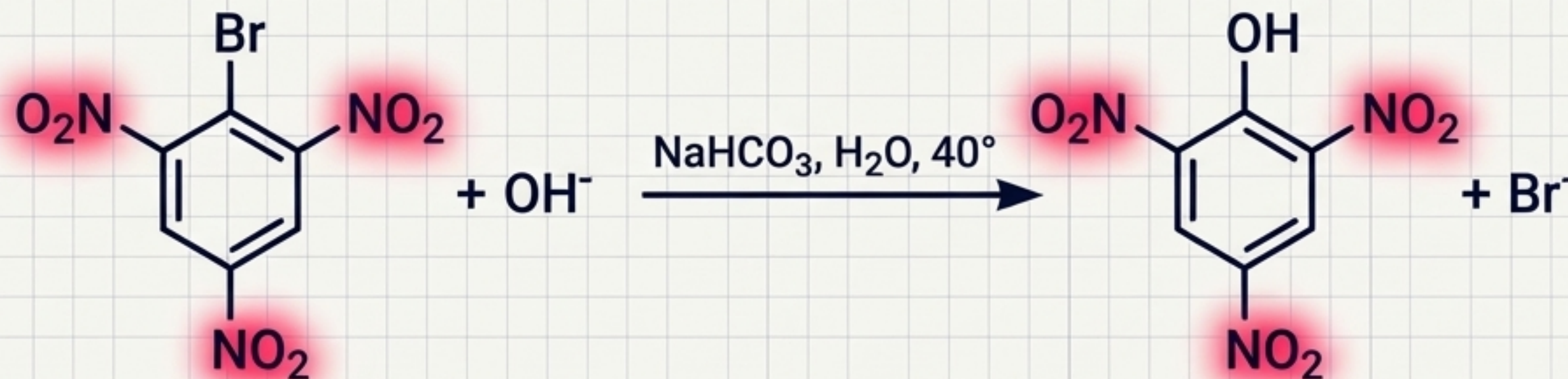
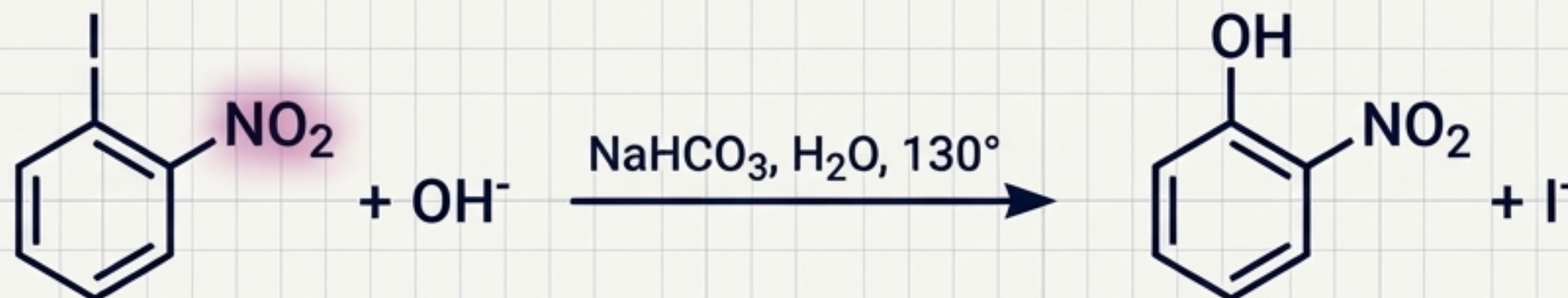


Path 2: The Sandmeyer Reaction



Breaching the Fortress

Nucleophilic substitution is only possible if the aryl halide is equipped with a 'Trojan Horse'—strong electron-withdrawing deactivators positioned precisely at the ortho or para locations to make the carbon susceptible to nucleophilic attack.



**Maximum
Withdrawing Power
Creates
Vulnerability:**
Reaction occurs at
dramatically lower
temperatures.

The Master Diagnostic Architecture

From first principles to practical application: analyze the competing forces to predict the substitution outcome.

